a Co-expression programmes
15 WGCNA modules: size, top set, hub proteins
Module
no set at FDR < 0.05 (turquoise)
PPP2R1A, UBE2L3, GOSR2
no set at FDR < 0.05 (blue)
CLIC5, MYLK, NIF3L1
Aerobic respiration and respira… (brown)
SAMM50, TIMM13, DLAT
no set at FDR < 0.05 (yellow)
ADPRHL1, CSTB, REEP5
L amino acid metabolic process (green)
IMPA1, TACO1, MFN2
Eukaryotic translation initiation (red)
RPLP0, RPLP2, RPL30
RHO GTPase effectors (black)
AHNAK, CLTC, HSPA5
Striated muscle contraction (pink)
NEB, TTN, TPM3
Monocarboxylic acid catabolic p… (magenta)
ETFDH, HSD17B8, SIRT5
mRNA metabolic process (purple)
RRAD, MPO, CTSG
Cell adhesion (greenyellow)
TAGLN, LUM, DSTN
Carbohydrate catabolic process (tan)
PGAM2, PLCD4, PHKA1
Cytoskeleton organization (salmon)
ANXA2, TUBB2A, PDIA3
no set at FDR < 0.05 (cyan)
CALR, CAPN2, PCBP1
Respiratory electron transport (midnightblue)
NDUFS1, CHCHD3, NDUFV1
0
100
200
proteins
b
Programmes across the nine contrasts
HR minus LR for between-group and interaction columns
Training
Acute
Between groups
Interaction
HR
LR
HR
LR
T1
T2
T3
Training
Acute
eigengene logFC
0.2
0.1
0.0
-0.1
-0.2
eigengene p < 0.05
fry p < 0.05
c Gene sets per contrast, fgsea and fry
fgsea ranks a set against the other proteins; fry tests its change across people.
1,378 sets; dashed: 69 fry calls expected by chance
fgsea FDR < 0.05
fry FDR < 0.05
fry p < 0.05
Training
HR
107
0
66
LR
116
0
37
Acute
HR
299
0
233
LR
88
0
94
Between groups
T1
70
0
13
T2
0
0
20
T3
229
0
193
Interaction
Training
64
0
26
Acute
206
0
140
Up
Down
d Cytoskeleton organization (salmon), person by person
Training Interaction logFC -0.13, p = 0.223; Acute Interaction logFC 0.24, p = 0.037
Thin lines: people; thick: group mean
HR
LR
eigengene
0.4
0.2
0.0
-0.2
T1
T2
T3
T1
T2
T3